NeuroSignalLab

Subject-Independent Motor-Imagery EEG Decoding with Classical and Deep-Learning Models

Abstract

Motor-imagery electroencephalography (EEG) is widely used as a tool to decode brain signals in brain-computer interface (BCI) research. However, reliable generalization across individuals remains difficult with EEG signals. Signals vary noticeably between subjects and are sensitive to preprocessing and evaluation design. Therefore, creating a versatile generalization tool becomes difficult [3,10,11]. NeuroSignalLab is a project made to understand these problems, and how different methods can mitigate these differences. The project investigates how reliably machine learning and deep-learning models can distinguish imagined left- versus right-fist movement in unseen subjects. The PhysioNet EEG Motor Movement/Imagery Dataset (EEGMMIDB) was used [1,2]. Recordings from 109 subjects were preprocessed into 4,898 trials using 64 EEG channels, common-average referencing, 7–30 Hz FIR filtering, and a +1 to +4 s analysis window. Subjects were separated into training, validation, and test cohorts containing 76, 16, and 17 subjects, respectively.

Five models were compared under the same development protocol. The five models were: ´CSP + LDA´, a generic convolutional neural network (CNN), ´ShallowConvNet´, ´EEGNet Configuration D´, and a max-norm-constrained EEGNet. The generic CNN obtained 0.530 ± 0.014 validation balanced accuracy (BA), CSP + LDA 0.573, ShallowConvNet 0.621 ± 0.007, Configuration D 0.628 ± 0.002, and constrained EEGNet 0.634 ± 0.012. Configuration D was selected as the primary model because its mean performance was competitive while its between-seed variability was lower than the constrained variant. A predetermined seed-42 checkpoint was then locked and used to evaluate the test cohort once. It achieved 57.6% BA and 57.2% macro F1 across 757 trials from 17 test subjects. Subject-level BA ranged from 45.8% to 80.1%, demonstrating substantial inter-subject variability.

Interpretability analyses showed class-dependent sensitivity around centro-parietal electrodes, with stronger occlusion effects around CP3/C3 for correct left-imagery predictions and CP4/C4 for correct right-imagery predictions. The first-layer EEGNet kernels also showed different frequency sensitivities, with peak normalized magnitudes approximately between 12 and 28 Hz within the 7–30 Hz input band. These findings characterize model behavior, not the neurophysiology. Overall, the project shows that EEG-specific convolutional neural networks can provide more reliable cross-subject decoding than a generic CNN under the same development protocol. The final test evaluation emphasized that inter-subject variability remained a major limitation.

1. Introduction

EEG provides a non-invasive method to measure recorded electrical activity on the scalp. Because each electrode measures mixtures of activity from multiple neural and non-neural sources, decoding requires careful preprocessing, experimental design, and interpretation. Motor imagery provides a controlled setting where imagining movement can alter sensorimotor rhythms where these decoding choices become easier to make. Event-related desynchronization and synchronization in frequency ranges commonly associated with mu and beta activity provide a baseline for where motor imagery can be found and processed [3].

A decoding classifier is only one component of a BCI system [2]. In an offline decoding project, like NeuroSignalLab, the central machine-learning question is therefore narrower: can a model learn patterns that distinguish tasks in recorded EEG, and can those patterns generalize to people whose data were never seen during training?

This distinction is important because EEG classification can be misleading during evaluation. Randomly splitting trials of the same participants into training and evaluation sets can allow subject-specific structure to appear on both sides of the split. Cross-subject evaluation is harder, but more directly tests whether a decoder transfers to new individuals. Inter-subject variability is a limitation in sensorimotor BCI research that the project explores thoroughly [10,11].

NeuroSignalLab was designed around two questions:

1) How reliably can classical machine-learning and EEG-specific deep-learning models decode left-versus-right motor imagery from open EEG recordings?
2) What do preprocessing, subject-independent evaluation, and analysis reveal about the limits of EEG machine learning?

2. Methods

2.1 Dataset and task

NeuroSignalLab uses the PhysioNet EEG Motor Movement/Imagery Dataset (EEGMMIDB), collected with the BCI2000 system [1,2]. EEGMMIDB contains 64-channel EEG recordings from motor execution and motor imagery tasks. The present analysis uses three repeated left-versus-right motor-imagery runs: runs 4, 8, and 12. In these runs, T1 corresponds to imagined left-fist movement and T2 to imagined right-fist movement. Rest (T0) was not used as a classification class.

The final processed dataset contains 109 subjects and 4,898 left/right motor-imagery trials.

2.2 Preprocessing

EEG preprocessing was implemented with MNE-Python [9]. All 64 EEG channels were retained. Channel names and sensor locations were standardized, and the recordings were common-average referenced. A 7–30 Hz FIR band-pass filter was applied to preserve the sensorimotor frequency range, while attenuating both slower and higher-frequency activity outside the target beta and mu bands.

Task epochs were extracted from -1 to +4 s relative to the motor-imagery event. The machine-learning input was restricted to +1 to +4 s to reduce reliance on the cue-evoked response. No voltage baseline correction was applied to the classifier inputs. At 160 Hz, the final model input for each trial was therefore: 64 channels × 481 samples. The saved EEG arrays remained in volts, while neural-network inputs were rescaled to microvolts during loading.

After preprocessing, the data was quality checked for channel structure, sampling frequency, event labels, trial counts, missing values, extreme amplitudes, and flat signals. Six subjects were previously reported as abnormal in an analysis of EEGMMIDB: S088, S089, S092, S100, S104, and S106. They were flagged rather than excluded [8]. Subject 88 required sampling-rate harmonization before event extraction. No participant was removed based on model performance, and no epoch-rejection threshold was introduced.

2.3 Subject-independent split

To avoid subjects leaking into multiple splits, subjects rather than trials were assigned to the development split using a fixed seed of 42:

Table 1. Subject-level train/validation/test split.

Split	Subjects	Trials
Training	76	3,423
Validation	16	718
Test	17	757
Total	**109**	**4,898**

The validation cohort was used for architecture, optimization, regularization, analysis, and final model selection. The 17-subject test cohort remained sealed throughout these decisions and was opened only after the primary checkpoint had been frozen.

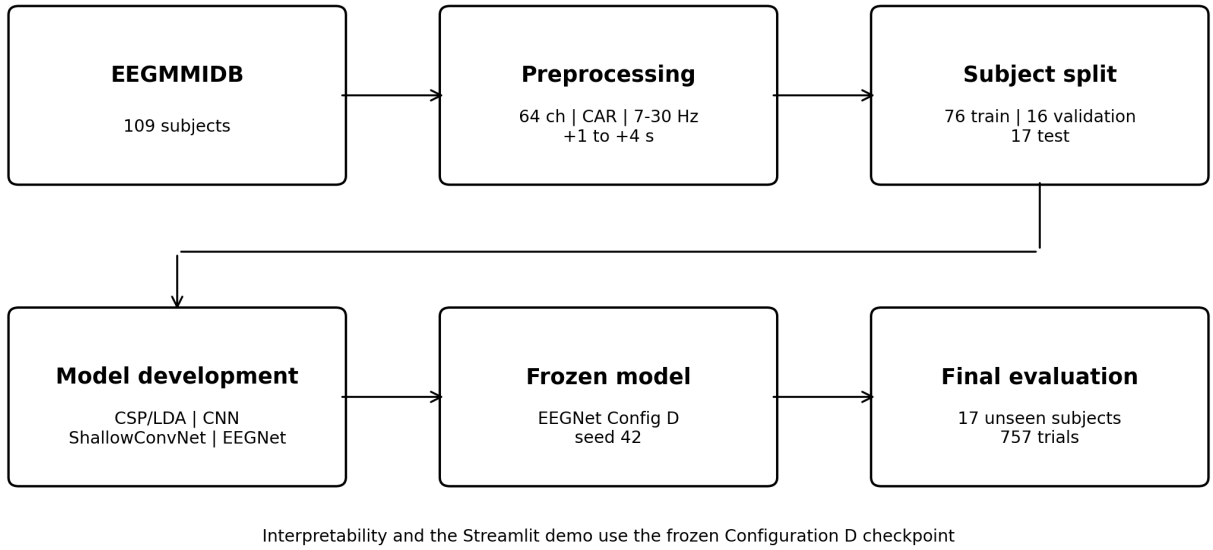


Figure 1. NeuroSignalLab analysis progression. The subject split remained during model development, and the Configuration D checkpoint was used for final evaluation, interpretability, and the Streamlit demo. Figure made using GPT 5.6 Sol.

2.4 Models

Four model families and five configurations were evaluated.

CSP + LDA. Common Spatial Patterns (CSP) learns spatial filters that maximize variance differences between classes, and is a well-established approach for imagined-movement EEG [4]. CSP features were classified using Linear Discriminant Analysis. Since common-average referencing makes one channel-space direction effectively redundant, CSP was fitted with an EEG rank of 63 instead of 64.

Generic CNN. A simple one-dimensional CNN was used as a deep-learning baseline. It consisted of two temporal Conv1d layers with ReLU activations and max pooling, followed by a linear classifier. Unlike EEG-specific architectures, it did not separate temporal and spatial filtering.

ShallowConvNet. ShallowConvNet was implemented using the Braindecode ShallowFBCSPNet architecture. This architecture was designed to learn temporal and spatial filters followed by operations that produce "power-like" representations and has previously been studied for EEG decoding [5].

EEGNet. EEGNet is an architecture designed specifically for EEG-based BCI decoding [6]. The NeuroSignalLab implementation uses temporal convolution, depthwise spatial convolution across the 64 electrodes, separable temporal convolution, average pooling, dropout, and a final linear classifier.

Table 2. Settings for EEGNet Configuration D

Configurations	Value/Variation
Optimizer	Adam
Loss	Cross-entropy
Learning rate	$1 * 10^{-3}$
Dropout	0.50
Weight decay	$1 * 10^{-4}$
Batch size	32
Maximum epochs	30
Max-norm	None

A secondary constrained EEGNet used max-norm limits of 1.0 on the depthwise spatial filters and 0.25 on the classifier weights, but retained the same experimental framework.

2.5 Reproducibility and model selection

Neural-network results were evaluated across five predefined random seeds: 42, 123, 456, 789, 2026.

This was intended to separate stable behavior from a lucky performance caused by random initialization, batch order, or dropout sequence. Balanced accuracy on the 16-subject validation cohort was the main selection metric.

Configuration D was selected both due to its high mean performance and lower variation across seeds. Configuration D was trained on seed 42 and set as a checkpoint before the final evaluation. The seed was chosen in advance. Its best validation checkpoint occurred at epoch 19 with a balanced accuracy of 0.6277.

No training/validation refit was performed before final testing. This preserved consistency between the checkpoint used for inference, interpretability, Streamlit demo, and the final test evaluation.

2.6 Evaluation and interpretability

The primary evaluation metric was balanced accuracy. Accuracy and macro F1 were also reported, together with per-subject BA and a confusion matrix.

Interpretability was applied to the frozen Configuration D model. Three methods were implemented: gradient saliency, channel occlusion and temporal occlusion.

Channel occlusion set one EEG channel to zero and measured the change in the target-class logit. For the group-level analysis, only correctly classified validation trials were used. Scores were first averaged within each subject and then across subjects so that subjects with more usable trials could not dominate the result.

Gradient saliency was also compared with saliency generated by a randomly initialized EEGNet, as a basic sanity check inspired by Adebayo et al. [7]. This is not a complete formal interpretability validation.

Finally, the eight kernels of EEGNet's first temporal convolution were inspected. Their 40 learned weights span 250 ms at 160 Hz. Fast Fourier transforms were used to visualize normalized magnitude-frequency responses. Zero-padding was used for smooth visualization.

2.7 Code and reproducibility

The project repository separates reusable code from experimental analysis. Finalized preprocessing, model, inference, and interpretability code are stored in src/neurosignallab/, while experimental analyses remain in notebooks/. The main implementation files are:

- *src/neurosignallab/preprocessing.py*
- *src/neurosignallab/models.py*
- *src/neurosignallab/inference.py*
- *src/neurosignallab/interpretability.py*
- *app/streamlit_app.py*

Saved subject splits, model configurations, metrics, predictions, and figures are kept in the project results and figures directories, so that the analyses can be inspected and reproduced. The raw PhysioNet recordings are not redistributed in the repository and must be obtained from the original dataset source [1].

3. Results

3.1 Evaluation design

Before the train/validation/test comparison, CSP + LDA was used to test how evaluation design affected performance.

Within-subject evaluation used Leave-One-Group-Out cross-validation per run and achieved a mean BA of 0.5973 with a 95% confidence interval of [0.5739, 0.6242]. Fixed cross-subject evaluation used Leave-One-Subject-Out cross-validation and achieved a mean BA of 0.56545 with a 95% confidence interval of [0.5476, 0.5869]. The mean difference from within-subject to fixed cross-subject evaluation was -0.0319 BA ($p = 0.0044$).

Nested cross-subject tuning used an outer Leave-One-Subject-Out loop with grouped inner cross-validation. It was mainly used to tune hyperparameters, and achieved a mean BA of 0.57531 with a 95% confidence interval of [0.5544, 0.5991]. Relative to the fixed cross-subject result, the mean increase was +0.0099 BA (p = 0.1582).

These earlier experiments used a different evaluation setup from the later model comparison and were therefore not placed in the same model-comparison table.

3.2 Development model comparison

Table 3. Validation model comparison.

Model	Validation balanced accuracy
Generic CNN	0.530 ± 0.014
CSP + LDA	0.573
ShallowConvNet	0.621 ± 0.007
EEGNet Configuration D	**0.628 ± 0.002**
Constrained EEGNet	0.634 ± 0.012

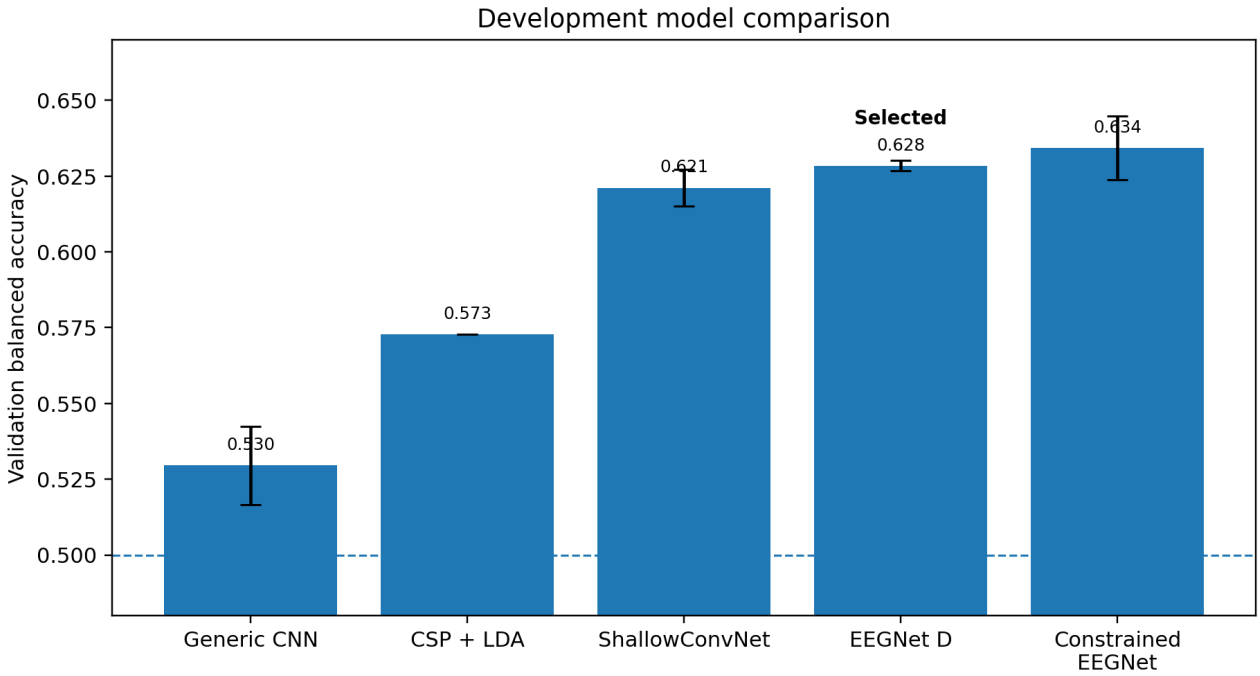


Figure 2. Development model comparison on the validation cohort. Error bars show standard deviation across five seeds for the neural-network models; CSP + LDA is the fixed-split result. Configuration D was selected for its combination of performance and seed stability.

The generic CNN reached approximately 0.99 training balanced accuracy while five-seed validation performance remained approximately 0.53.

The constrained EEGNet achieved the highest mean validation BA, 0.6342, compared with 0.6284 for Configuration D. Its between-seed standard deviation was 0.0118 compared with 0.0019 for Configuration D. The mean difference between the two models was approximately 0.006 BA. Their subject-level performance was highly correlated (r = 0.971), and the constrained model improved some validation subjects while worsening others.

Configuration D was selected as the primary model.

3.3 Final test evaluation

The Configuration D checkpoint trained on seed 42 was evaluated once on 757 trials from 17 previously unseen subjects.

Table 4. Final held-out test metrics.

Metric	Final test result
Accuracy	0.5760
Balanced accuracy	0.5756

Metric	Final test result
Macro F1	0.5721
Mean subject BA	0.5776
Median subject BA	0.5810
Subject BA range	0.458–0.801

The final confusion matrix was:

Table 5. Final held-out confusion matrix.

	Predicted left	Predicted right
True left	254	126
True right	195	182

Left-class recall was approximately 66.8%, while right-class recall was approximately 48.3%. The test cohort contained 380 left and 377 right trials. The model produced 449 left predictions and 308 right predictions.

The frozen checkpoint's validation BA was 0.6277, compared with 0.5756 on the final test set, a decrease of approximately 5.2 percentage points.

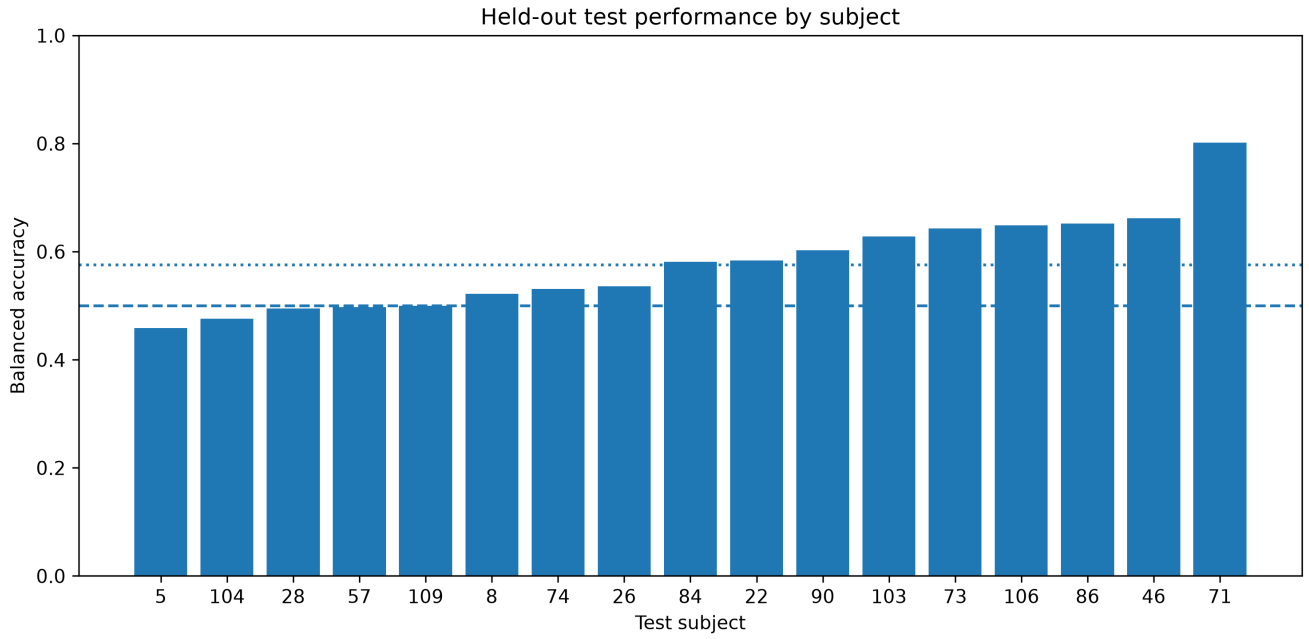


Figure 3. Balanced accuracy for each of the 17 test subjects. The dashed line marks 0.50 balanced accuracy as a descriptive reference. The dotted line marks the mean BA.

Twelve of 17 subjects had BA above 0.50, one had BA equal to 0.50, and four had BA below 0.50. The strongest subject reached approximately 0.80 BA, while the weakest reached approximately 0.46 BA.

No statistical test against chance was performed on the final cohort. The report therefore presents the observed performance directly rather than making a formal claim of decoding above chance.

3.4 Interpretability

Channel occlusion produced an asymmetrical class-dependent pattern. For correctly classified left-imagery trials, the largest mean target-logit decreases occurred around CP3, C3, P3, and CP1. For correctly classified right-imagery trials, the strongest effects occurred around CP4, C4, CP6, and P4.

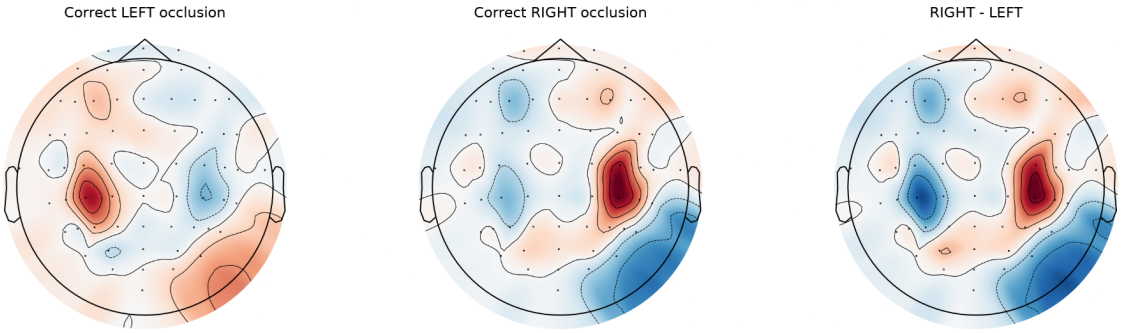


Figure 4. Subject-balanced channel-occlusion topomaps for correctly classified validation trials. The maps show model sensitivity at the electrode level.

The limited saliency sanity check compared ten validation trials between the trained and a randomly initialized network. Correlations between trained and random saliency maps had a mean of approximately -0.13, with values ranging from approximately -0.20 to -0.02 [7].

The eight first-layer temporal kernels showed different frequency responses. Within the 7–30 Hz input band, their peak normalized magnitude responses occurred approximately between 12 and 28 Hz. The kernels span 250 ms.

4. Discussion

4.1 Evaluation design affected the apparent performance

The within-subject CSP + LDA result was higher than the fixed cross-subject result. This supports the decision to keep subjects separated when the goal is to test generalization to new people. Randomly or more easily evaluating data from the same participants can produce a more optimistic estimate than cross-subject evaluation.

Nested tuning increased mean cross-subject BA from 0.56545 to 0.57531, but the difference was not statistically significant ($p = 0.1582$). In this project, changing the evaluation design had a clearer effect than the tuning.

The earlier CSP experiments were also not directly compared numerically with the later neural-network experiments.

4.2 EEG-specific representation improved development performance

The central result was not simply that deeper models outperformed simpler models. The generic CNN could fit the training population extremely well but transferred poorly to the validation subjects. The large training-validation gap indicates overfitting and poor transfer to new subjects.

By contrast, ShallowConvNet and EEGNet incorporated structures specifically suited to multichannel EEG, including separate temporal and spatial operations [5,6], and both produced substantially stronger cross-subject validation performance.

This result is limited to the implementations and protocol tested here. It does not establish that EEGNet is universally superior to CSP, generic CNNs, or other EEG models. It does show that model choice affected generalization within the project.

4.3 Reproducibility mattered more than the single highest score

The constrained EEGNet produced the highest mean validation BA but had greater variation across random seeds. Selecting it solely because $0.634 > 0.628$ would ignore uncertainty introduced by training.

Configuration D instead showed nearly identical performance across five seeds. This stability was particularly important because the project's main goal was reliable modeling, rather than maximizing one development number. The result also illustrates why a single neural-network run is insufficient to characterize model performance.

Configuration D was selected as the primary model because it combined competitive performance with greater consistency.

4.4 Inter-subject variability remained the dominant limitation

The final test BA of 57.6% was lower than the frozen checkpoint's 62.8% validation BA, and performance ranged from 45.8% to 80.1% across unseen subjects. This variation is consistent with the broader literature, where differences in EEG feature distributions between individuals are recognized obstacles. [10,11].

The development analysis reinforced the same pattern. Subject-level performance was highly correlated between Configuration D and constrained EEGNet ($r = 0.971$), and also strongly correlated between Configuration D and ShallowConvNet ($r \approx 0.908$). This suggests that some validation subjects were consistently easier or harder to decode across models. However, no model was best for every subject, so the limitation cannot be reduced to just architecture alone or subject variability alone.

Future work could investigate subject-specific calibration, transfer learning, domain adaptation, session-to-session robustness, or external validation on independent datasets. These possibilities remain future research directions.

4.5 Interpretability describes the model, not the brain

The lateralized occlusion result is one of the most interpretable findings in the project. The pattern is consistent with the model using different electrode-level information for the two output classes. Even so, it only gives evidence about model sensitivity. It cannot say anything about physiological causality with certainty, since scalp EEG records mixtures of different signal sources. Channel zeroing is also an artificial perturbation, and common-average referencing means that the resulting occluded input no longer preserves the same relationships as the original EEG.

The limited saliency sanity check indicates that the trained parameters affected the observed saliency patterns, but the interpretability is not fully validated [7].

The same caution applies to the learned temporal kernels. Their frequency-response peaks overlap ranges relevant to sensorimotor EEG, but frequency overlap is not sufficient to prove that kernels correspond to physiological rhythms. The model was optimized for classification, not for source identification. Because the kernels span only 250 ms and later layers combine these features, individual filters were not assigned specific physiological identities such as “mu” or “beta” filters.

4.6 Clinical and BCI scope

NeuroSignalLab is an offline EEG-decoding research prototype. It does not include live EEG acquisition, continuous real-time preprocessing, online artifact handling, adaptive calibration, decision smoothing, external-device control, or closed-loop feedback.

The project therefore implements only the decoding component that could exist within a larger BCI pipeline. It has not evaluated patients, rehabilitation outcomes, or neuroplasticity within patients. The results should not be interpreted as evidence of a clinically deployable BCI, diagnostic system, or medical device.

Additional limitations include reliance on one public dataset, a binary cue-based motor-imagery task, no independent external dataset, no dedicated probability calibration, and final test evaluation of only a single checkpoint. The development analysis supports reproducibility, but it does not prove that the final test score itself is reproducible across independently trained models.

5. Conclusion

NeuroSignalLab developed a pipeline for left-versus-right motor-imagery EEG classification, progressing from preprocessing and classical spatial filtering to generic and EEG-specific neural networks, and later multi-seed model selection, evaluation, interpretability, and an interactive interface.

Under the development protocol, EEG-specific neural architectures generalized more reliably than the generic CNN. Configuration D EEGNet was selected because it combined competitive performance with stability across random seeds. When the seed-42 model was evaluated on 17 previously unseen test subjects, it achieved 57.6% balanced accuracy, with substantial variation between individuals and asymmetric left/right recall.

The project demonstrates how strongly conclusions in EEG machine learning depend on independent splitting, reproducibility analysis, a held-out final cohort, and careful interpretation. EEGNet learned

structured temporal and electrode-level sensitivities, but these results describe the behavior of an offline decoder rather than the neurophysiology itself.

References

[1] G. Schalk, **“EEG Motor Movement/Imagery Dataset,”** PhysioNet, Version 1.0.0, 2009. doi: **10.13026/C28G6P**.

[2] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, **“BCI2000: A General-Purpose Brain-Computer Interface (BCI) System,”** *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004. doi: **10.1109/TBME.2004.827072**.

[3] G. Pfurtscheller and F. H. Lopes da Silva, **“Event-related EEG/MEG synchronization and desynchronization: basic principles,”** *Clinical Neurophysiology*, vol. 110, no. 11, pp. 1842–1857, 1999. doi: **10.1016/S1388-2457(99)00141-8**.

[4] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, **“Optimal spatial filtering of single trial EEG during imagined hand movement,”** *IEEE Transactions on Rehabilitation Engineering*, vol. 8, no. 4, pp. 441–446, 2000. doi: **10.1109/86.895946**.

[5] R. T. Schirrmeister et al., **“Deep learning with convolutional neural networks for EEG decoding and visualization,”** *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017. doi: **10.1002/hbm.23730**.

[6] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, **“EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces,”** *Journal of Neural Engineering*, vol. 15, no. 5, 056013, 2018. doi: **10.1088/1741-2552/aace8c**.

[7] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, **“Sanity Checks for Saliency Maps,”** in *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018.

[8] Z. Shuqfa, A. Lakas, and A. N. Belkacem, **“Increasing accessibility to a large brain-computer interface dataset: Curation of PhysioNet EEG Motor Movement/Imagery Dataset for decoding and classification,”** *Data in Brief*, vol. 54, 110181, 2024. doi: **10.1016/j.dib.2024.110181**.

[9] A. Gramfort et al., **“MEG and EEG data analysis with MNE-Python,”** *Frontiers in Neuroscience*, vol. 7, 267, 2013. doi: **10.3389/fnins.2013.00267**.

[10] S. Saha and M. Baumert, **“Intra- and Inter-subject Variability in EEG-Based Sensorimotor Brain Computer Interface: A Review,”** *Frontiers in Computational Neuroscience*, vol. 13, 87, 2020. doi: **10.3389/fncom.2019.00087**.

[11] M. Ahn and S. C. Jun, **“Performance variation in motor imagery brain-computer interface: A brief review,”** *Journal of Neuroscience Methods*, vol. 243, pp. 103–110, 2015. doi: **10.1016/j.jneumeth.2015.01.033**.